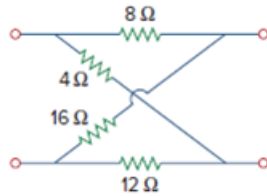


Problem

Determine the z and y parameters for the circuit in Fig. 19.78.

Figure 19.78



Step-by-step solution

Step 1 of 9

Refer to the Figure 19.78 in the textbook.

The z parameters are describe as,

$$V_1 = z_{11}I_1 + z_{12}I_2$$

$$V_2 = z_{21}I_1 + z_{22}I_2$$

$$z_{11} = \left. \frac{V_1}{I_1} \right|_{I_2=0}$$

$$z_{21} = \left. \frac{V_2}{I_1} \right|_{I_2=0}$$

$$z_{12} = \left. \frac{V_1}{I_2} \right|_{I_1=0}$$

$$z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1=0}$$

Step 2 of 9

To determine z_{11} and z_{21} , apply voltage V_1 to the port 1 and open port 2.

Thus, the circuit become as in Figure 1.

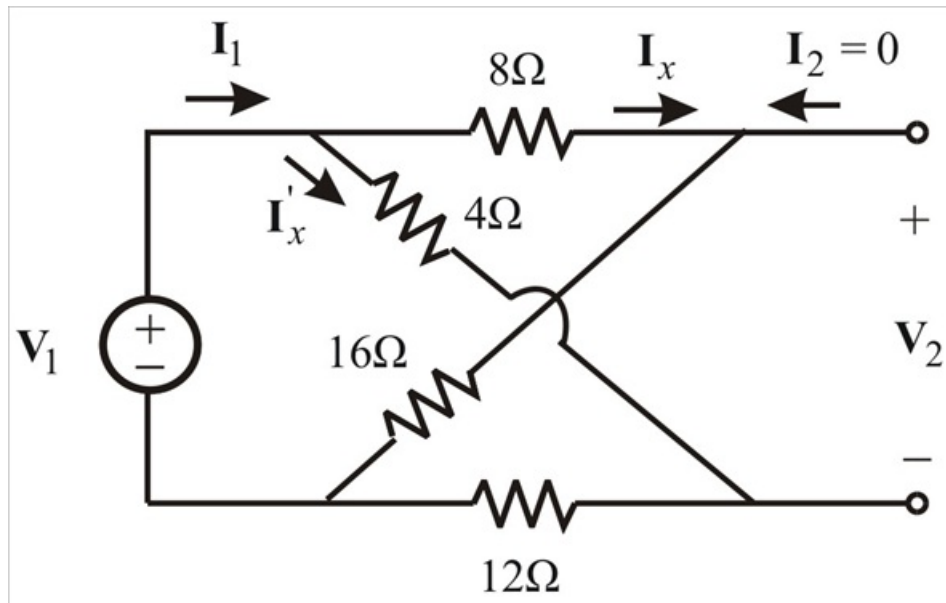


Figure 1

Step 3 of 9

From the above circuit same current flows through 8Ω resistor and 16Ω , so both are in series. Similarly, same current flows through 4Ω resistor and 12Ω , so both are in series.

The voltage at port 1 .

$$V_1 = I_1 [(4+12) \parallel (8+16)]$$

$$\frac{V_1}{I_1} = \frac{(16)(24)}{16+24}$$

$$= 9.6$$

$$z_{11} = \frac{V_1}{I_1}$$

$$= 9.6 \Omega$$

Step 4 of 9

Use current division rule.

$$\begin{aligned} I_x &= I_1 \left[\frac{12+4}{(12+4)+(8+16)} \right] \\ &= 0.4I_1 \end{aligned}$$

$$\begin{aligned} I'_x &= I_1 \left[\frac{16+8}{(12+4)+(8+16)} \right] \\ &= 0.6I_1 \end{aligned}$$

Apply KVL in the output loop.

$$-V_2 - 8I_x + 4I'_x = 0$$

$$-V_2 - 8(0.4)I_1 + 4(0.6)I_1 = 0$$

$$V_2 = -0.8I_1$$

$$\frac{V_2}{I_1} = -0.8$$

$$\begin{aligned} z_{21} &= \frac{V_2}{I_1} \\ &= -0.8 \Omega \end{aligned}$$

Step 5 of 9

To determine z_{12} and z_{22} , apply voltage V_2 to the port 2 and open port 1.

Thus, the circuit become as shown in Figure 2.

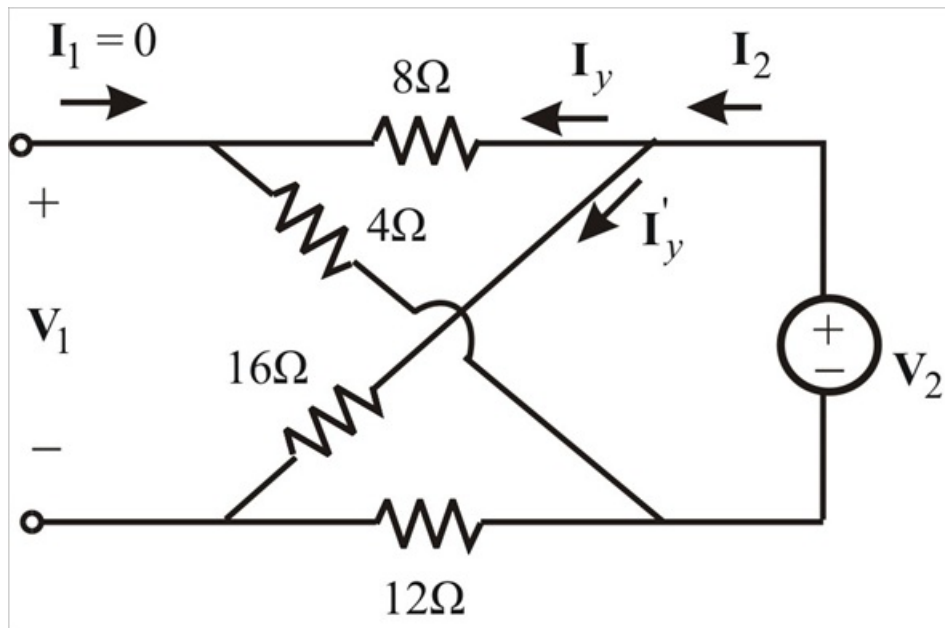


Figure 2

From the circuit in Figure 2 .

The voltage at port 2 .

$$V_2 = [(16+12) \parallel (8+4)] I_2$$

$$V_2 = (28 \parallel 12) I_2$$

$$\frac{V_2}{I_2} = 8.4$$

$$\begin{aligned} z_{22} &= \frac{V_2}{I_2} \\ &= 8.4 \Omega \end{aligned}$$

Step 6 of 9

Use current division rule.

$$\begin{aligned} \mathbf{I}_y &= \frac{16+12}{(16+12)+(8+4)} \mathbf{I}_2 \\ &= 0.7 \mathbf{I}_2 \end{aligned}$$

$$\begin{aligned} \mathbf{I}'_y &= \frac{8+4}{(16+12)+(8+4)} \mathbf{I}_2 \\ &= 0.3 \mathbf{I}_2 \end{aligned}$$

Apply Kirchhoff's voltage law at the port 1.

$$-\mathbf{V}_1 + 4\mathbf{I}_y - 12\mathbf{I}'_y = 0$$

$$-\mathbf{V}_1 + 4(0.7\mathbf{I}_1) - 12(0.3\mathbf{I}_1) = 0$$

$$-\mathbf{V}_1 + 2.8\mathbf{I}_y - 3.6\mathbf{I}_1 = 0$$

$$\mathbf{V}_1 = -0.8\mathbf{I}_2$$

$$\frac{\mathbf{V}_1}{\mathbf{I}_2} = -0.8$$

$$\begin{aligned} \mathbf{z}_{12} &= \frac{\mathbf{V}_1}{\mathbf{I}_2} \\ &= -0.8 \, \Omega \end{aligned}$$

Therefore, the $\mathbf{z}$ parameters for the circuit in Figure 19.78 is $\boxed{\begin{bmatrix} 9.6 & -0.8 \\ -0.8 & 8.4 \end{bmatrix} \, \Omega}$.

Step 7 of 9

To calculate $\mathbf{y}$ parameters, use conversion formula from $\mathbf{z}$ parameters to $\mathbf{y}$ parameters.

$$\mathbf{y} = \begin{bmatrix} \frac{z_{22}}{\Delta z} & -\frac{z_{12}}{\Delta z} \\ -\frac{z_{21}}{\Delta z} & \frac{z_{11}}{\Delta z} \end{bmatrix}$$

Here $\Delta z = z_{11}z_{22} - z_{12}z_{21}$.

Use the $\mathbf{z}$ parameter values to calculate Δz .

$$\begin{aligned} \Delta z &= 9.6(8.4) - 0.8(0.8) \\ &= 80.64 - 0.64 \\ &= 80 \end{aligned}$$

Step 8 of 9

Therefore, the $\mathbf{y}$ parameters are calculated as,

The parameter $\mathbf{y}_{11}$.

$$\begin{aligned} \mathbf{y}_{11} &= \frac{z_{22}}{\Delta z} \\ &= \frac{8.4}{80} \\ &= 0.105 \, \text{S} \end{aligned}$$

The parameter $\mathbf{y}_{12}$.

$$\begin{aligned} \mathbf{y}_{12} &= -\frac{z_{12}}{\Delta z} \\ &= -\frac{(-0.8)}{80} \\ &= 0.01 \, \text{S} \end{aligned}$$

The parameter y_{21} .

$$\begin{aligned} y_{21} &= -\frac{z_{21}}{\Delta z} \\ &= -\frac{(-0.8)}{80} \\ &= 0.01 \text{ S} \end{aligned}$$

The parameter y_{22} .

$$\begin{aligned} y_{22} &= \frac{z_{11}}{\Delta z} \\ &= \frac{9.6}{80} \\ &= 0.12 \text{ S} \end{aligned}$$

Therefore, the y parameters for the circuit in Figure 19.78 are $\begin{bmatrix} 0.105 & 0.01 \\ 0.01 & 0.12 \end{bmatrix} \text{ S}$.